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Competitive Coding Practical EST

1) Travelling Salesmen Problem

⇒ ~~Start from city 0.~~

⇒ Approach / Algorithm:

1. Start from city 0.
2. Use bitmask to represent visited cities.
3. Try going to all unvisited cities.
4. Recursively compute cost.
5. Add return cost to starting city.
6. Take minimum of all possibilities.

⇒ Test case:

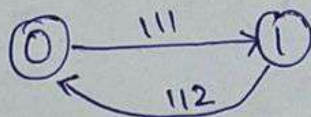
Input: $cost[i][j] = [[0, 11], [11, 2, 0]]$

Output: 223

⇒ Time Complexity: $O(n^2 \cdot 2^n)$

⇒ Space Complexity: $O(n \cdot 2^n)$

⇒ Graph:



⇒ So, beginning from city 0, we can only go to city

1, mark city 0 visited.

⇒ After visiting city 1, mark it visited and add its cost.

⇒ Then return to city 0 and add its cost.

⇒ As it has only 2 nodes/cities, answer becomes:

$$\text{ans} = 11 + 112$$

$$= 223$$

2) Prim's Algorithm :

→ Approach :

- 1) Initialize an ~~array~~ ^{min priority queue} to track minimum weighted edges.
- 2) Start from any vertex.
- 3) Mark it as visited.
- 4) Push all its edges into a min priority queue.
- 5) Repeat until all vertices are included :-
 - i) Pick the smallest weight edge.
 - ii) If the destination vertex is not visited,
 - a) Add it to MST
 - b) Mark visited
 - c) Push its edges into the queue.

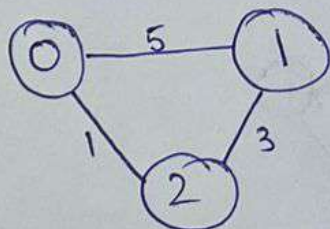
⇒ Test case :

$V=3$, $E=3$, Edges = $[[0,1,5], [1,2,3], [0,2,1]]$

→ Time Complexity : $O(E \log V)$

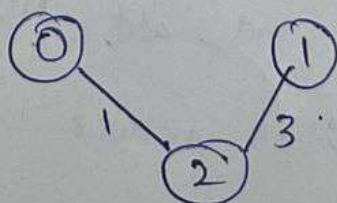
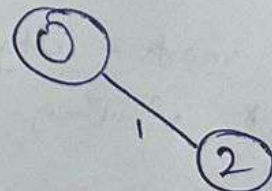
→ Space Complexity : $O(V+E)$

→ Graph :



⇒ If beginning from 0 node, choose next edge with minimum weight, so we choose 0 to 2

→ Then, same thing from node 2, we choose 1.



⇒ MST and total weight = 4